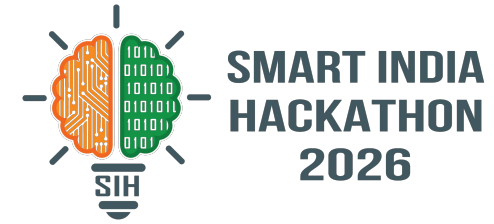
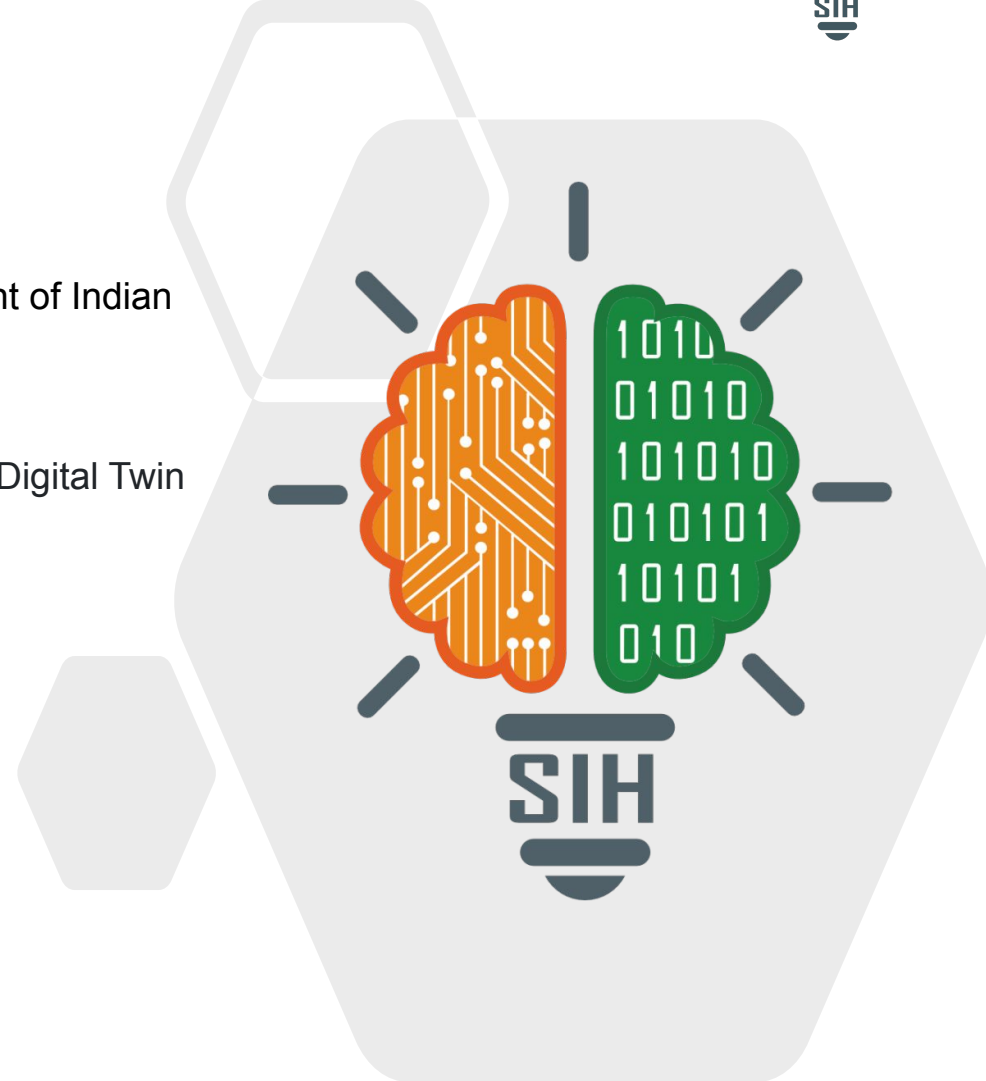


SMART INDIA HACKATHON 2026



- **Problem Statement ID** – 26060
- **Problem Statement Title** - Digital Platform for efficient remote management of Indian Antarctic Research Stations
- **Solution Title** – DhruvaTwin: A Pole-Star-Inspired, Resilient AIoT-Enabled Digital Twin Framework for Antarctic Research Stations
- **Theme** – Smart Automation
- **PS Category** - Software
- **Team ID** - 126510
- **Team Name** - HackFinity007
- **Demo Link** - <https://dhruva-twin-antarctic-stations-digi.vercel.app/>



Problem Addresses

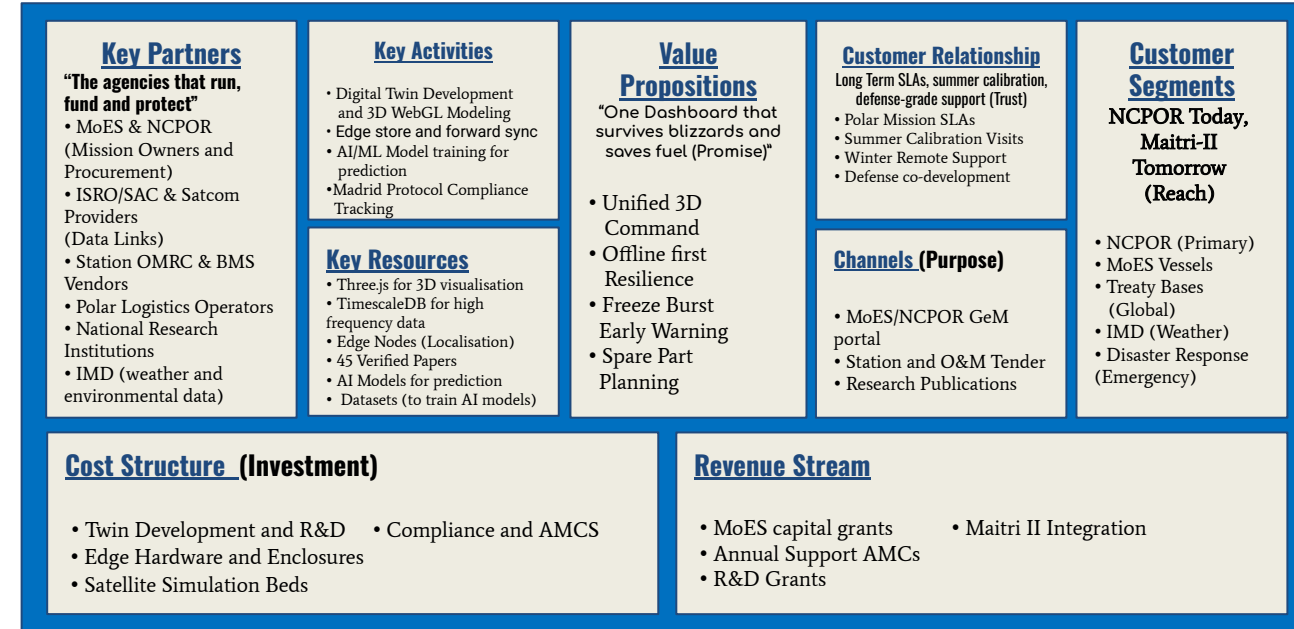
Station-Level Operational Threats :

- ➔ **Zero Unified Command:** No Digital Twin exists for Maitri or Bharati; NCPOR lacks a centralized 3D operational overview across both stations. (Source: NCPOR Station Records)
 - ➔ **Blizzard Telemetry Blackout:** Katabatic storms sever VSAT line-of-sight, leaving Command HQ blind during critical emergencies. (Source: 44th/45th ISEA Reports, NCPOR)
 - ➔ **45-Minute Freeze Trap:** Maitri's Lake Priyadarshini water pipeline ruptures in 45 minutes if electric trace heating trips. (Source: NCPOR Technical Documentation)
 - ➔ **Two Stations, Two Eras:** Maitri (1989) relies on analog relays and manual logs, while Bharati (2012) runs Modbus/BACnet buses with no shared twin. (Source: NCPOR Station Records)
 - ➔ **One Ship, One Year Resupply Trap:** Resupply arrives strictly once annually (Dec–Mar); mid-winter pump failures leave the station with zero spares for 6+ months. (Source: 45th ISEA Expedition Operations)
- Mission & Environmental Consequences :**
- ➔ **Fuel Viscosity Surge & Microgrid Freeze-Out:** Sub-zero Jet A-1 crystallization clogs generator lines, risking sudden shutdown and habitat heat loss. (Source: IEEE Antarctic Microgrids)
 - ➔ **Indian Antarctic Act & Treaty Violations:** Inadequate auditing of emissions, fuel leaks, and wastewater effluents risks severe domestic and international penalties. (Source: Indian Antarctic Act, 2022 / Madrid Protocol)
 - ➔ **Global Polar Telemetry Gaps:** Power and uplink outages cause unrecoverable gaps in climate and ozone monitoring submitted to global bodies. (Source: NCPOR Polar Science Telemetry Data)

Idea / Approach Details

- ➔ **Unified Station Intelligence:** Integrates power, thermal, water, equipment & environmental data from Maitri & Bharati.
- ➔ **AIoT Ingestion:** Collects, filters & normalizes telemetry via REST/MQTT into an NPDC-ready schema.
- ➔ **Edge AI & Physics:** Enables anomaly detection, energy forecasting & freeze-risk prediction.
- ➔ **Offline-First Sync:** Uses local buffering, priority queuing & delta sync during connectivity loss.
- ➔ **Self-Recovery & Security:** Provides FPGA-based fault recovery with secure encrypted communication.
- ➔ **Living Digital Twin:** WebGL/Three.js enables what-if simulation, spares forecasting & remote decisions.

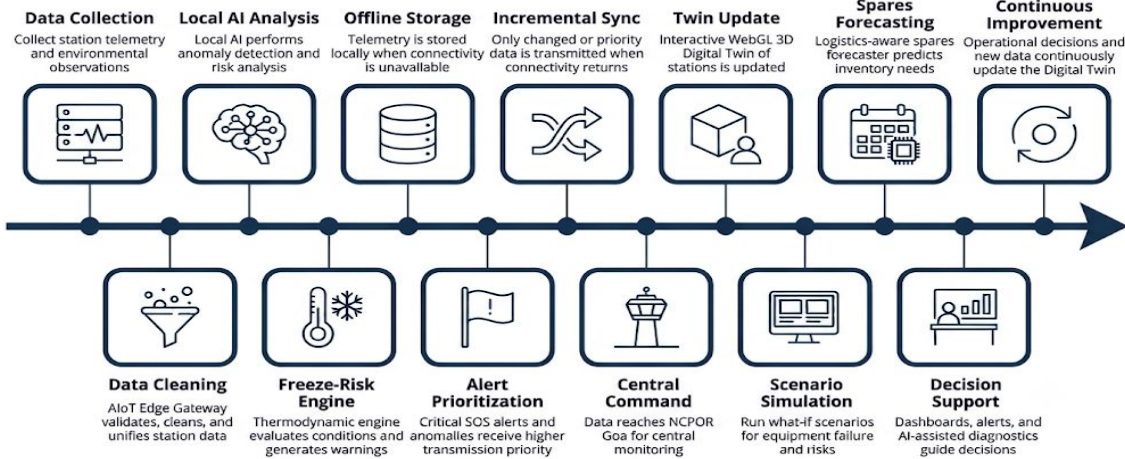
BUSINESS MODEL



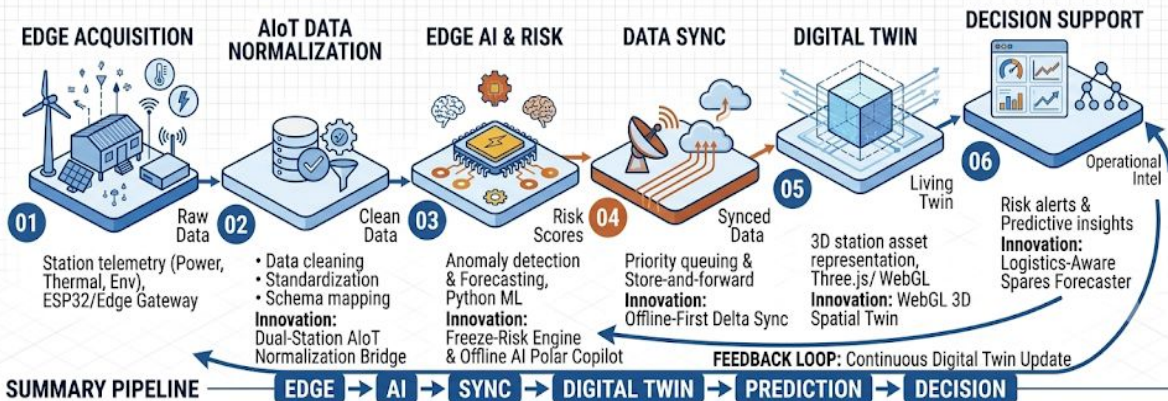
Innovation & Uniqueness

- ➔ **WebGL 3D Spatial Twin:** Zero-install 3D visualization and what-if operational simulation.
- ➔ **Thermodynamic Freeze-Risk Engine:** Predicts water-system freeze risk and provides early operational warnings.
- ➔ **Offline-First Delta Sync:** Store-and-forward edge caching with priority-based synchronization.
- ➔ **Dual-Station AIoT Normalization Bridge:** Intelligent edge nodes normalize multi-station telemetry and perform localized anomaly filtering into an NPDC-ready schema.
- ➔ **Logistics-Aware Spares Forecaster:** Links equipment health and spare requirements with inventory and resupply constraints.
- ➔ **FPGA-Based Edge Self-Recovery:** Enables rapid sensor-fault detection and modular recovery at the edge.
- ➔ **Offline AI Polar Copilot:** Local RAG assistant for diagnostics, SOP retrieval, and compliance support.

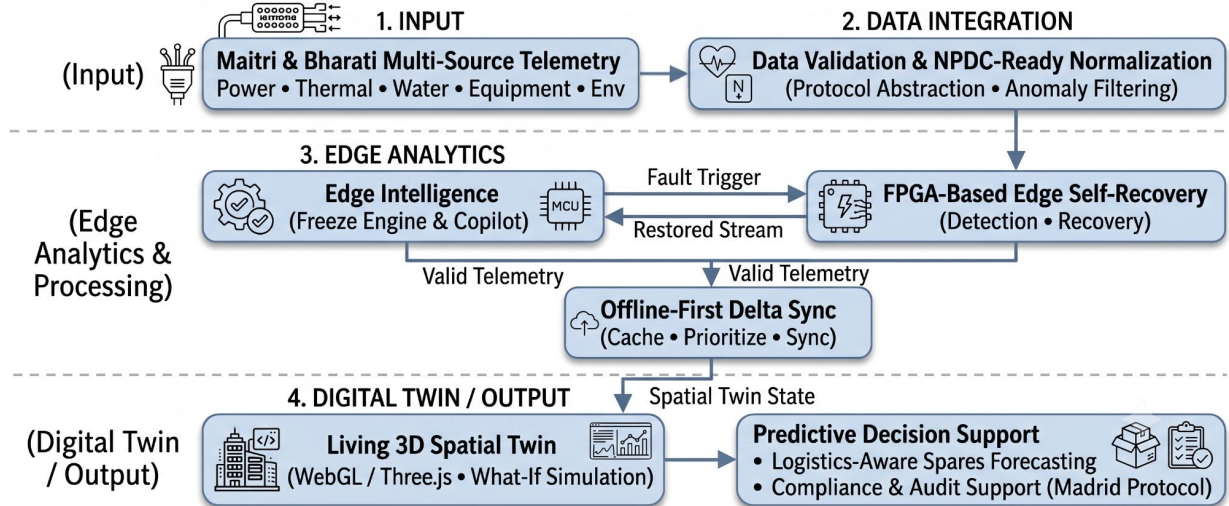
DhruvaTwin — End-to-End Operational Process Flow



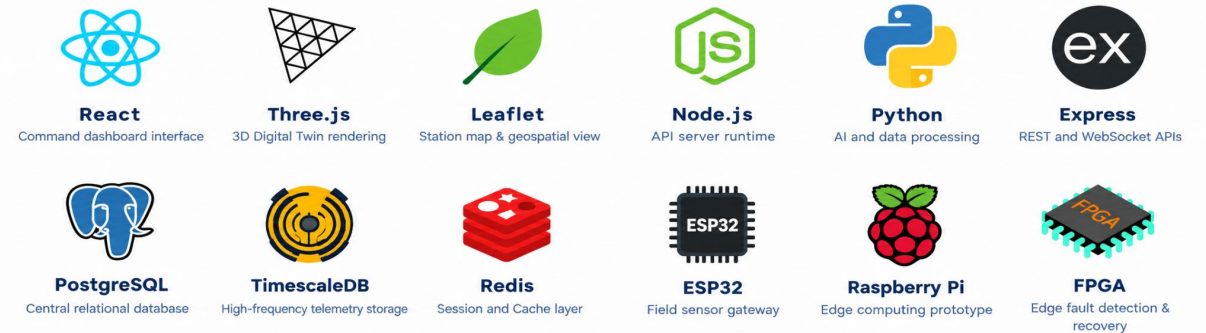
DhruvaTwin — Model Implementation



DhruvaTwin — Model Explanation



DhruvaTwin — Tech Stack



FEASIBILITY & VIABILITY



Technical Feasibility



Rugged Modular Architecture
React/FastAPI stack powered by TimescaleDB, with wide-temp (-40°C) edge gateways running local SQLite ring buffers.



Silicon-Level Reliability
FPGA hardware watchdog with Triple Modular Redundancy (TMR) detects sub-zero sensor drift and isolates faults without software reboots.



Zero-Install 3D Spatial Twin
Lightweight WebGL/Three.js CAD rendering delivers smooth 60 FPS performance on standard laptops at NCPOR HQ.



Physics + AI Edge Analytics
Solves transient heat-conduction PDEs for a 45-min freeze warning, coupled with a quantized offline AI Copilot.



Operational Feasibility



Zero-Risk Non-Invasive Tap
Uses opto-isolated, read-only interfaces ensuring zero operational interference with primary station life-support PLCs.



Offline-First Polar Autonomy
Full local operation during multi-week blizzard station blackout via store-and-forward edge queuing.



Human-in-the-Loop Decision Support
Advisory-only alerts and What-If failure simulations empower on-station engineers and NCPOR directors.



Domain-Aligned Logistics
Predictive Remaining Useful Life (RUL) models align directly with the annual Cape Town maritime resupply window.



Financial Viability



Open-Source Cost Avoidance
Eliminates recurring enterprise twin licenses by utilizing Python, Three.js, TimescaleDB, and Protobuf.



Brownfield Infrastructure Reuse
Ingests legacy Maitri analog lines and modern Bharati Modbus/BACnet networks without replacing existing sensors.



Extreme Satcom Bandwidth Savings
Protobuf delta sync achieves a 99.5% payload reduction (250 KB → 1.2 KB), minimizing costly polar satellite bandwidth.



High-Value Failure Prevention
Preventing a single water pipeline rupture or generator freezeout saves crores in emergency repairs and fuel logistics.



Scalability & Sustainability



Dual-Station to National Scale
Engineered for Maitri and Bharati, with immediate extensibility to the upcoming Maitri-II and Arctic base Himadri.



NPDC Open Science Ready
Ingested streams automatically normalize to FAIR-compliant schemas and NetCDF-4 archives for global scientific research.



Statutory Compliance Automation
Automated audit logging enforces mandatory reporting under the Indian Antarctic Act 2022 and the Madrid Protocol.



Containerized Microservices
Docker-based deployment enables Zero-touch remote patching and model updates over low-bandwidth data links.

Potential Challenges & Risks



Extreme Cold & Environment:

Harsh environmental conditions (-40°C)



Satellite Connectivity:

Intermittent satellite connectivity & drops



Heterogeneous Formats:

Diverse station systems & data schemas



Limited Historical Data:

Sparse operational and failure telemetry



Hardware & Sensor Faults:

Sensor drift and edge-hardware failures



Cybersecurity & Integrity:

Tampering risks & data-integrity threats



Existing Infrastructure:

Integration with legacy station PLCs



Prediction Errors & Noise:

Model false positives & alert fatigue



Strategies for Overcoming Challenges

Ruggedized Edge Hardware:

Modular hardware & local edge processing



Offline-First Architecture:

Buffering, priority queuing & delta sync



AIoT Normalization Bridge:

Schema mapping, validation & NPDC FAIR



Hybrid Data Strategy:

Real + simulated data progressive validation



Edge Fault Self-Recovery:

Edge fault detection, filtering & recovery



Zero-Trust Security & RBAC:

Authentication, encryption & audit logging



Non-Invasive API Integration:

Read-only tap & REST/MQTT interfaces



Human-in-the-Loop Validation:

Confidence thresholds + expert verification



Potential Impact on the Target Audience



➔ Unified remote visibility across Maitri & Bharati, improving situational awareness of critical station operations



➔ Earlier identification of operational risks, supporting proactive maintenance and reducing unexpected disruptions.



➔ Improved resilience during connectivity interruptions through offline-first data buffering and synchronization.



➔ Faster and better-informed decisions through Digital Twin visualization, predictive alerts and What-If simulations.



➔ Smarter resource & logistics planning, supporting energy management, equipment health and spare-part forecasting.

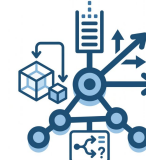
Benefits Of The Solution



➔ **Predictive risk detection** for earlier operational intervention.



➔ **Resilient data continuity** during connectivity disruptions.



➔ **Faster decisions** through Digital Twin insights and What-If simulation.



➔ **Smarter resource planning** for energy, maintenance and spare parts.



➔ **Scalable remote management** across Indian polar stations.

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Technical / Operational Area	Current Solution / Existing Ecosystem	DhruvaTwin — Proposed Framework
Unified Command	Siloed station systems	Unified 3D Digital Twin
Communication	Connectivity-dependent monitoring	Offline-First Delta Sync
Water Freeze Risk	Reactive/manual monitoring	Thermodynamic Freeze-Risk Engine
Fault Tolerance	Sensor drift & failures	FPGA-Based Self-Recovery
Emergency Support	Manual SOP lookups	Offline AI Polar Copilot
Logistics	Manual spare planning	Predictive Spares Forecasting
Compliance	Manual record keeping	Automated Audit Support
Data Standardization	Disparate data formats	AIoT Normalization Bridge
Safety Integration	Risk of invasive integration	Read-Only Fail-Safe Ingest
Innovation Scope	Domain-specific monitoring	Predictive, Integrated Digital Twin
Future Scalability	Station-specific solutions	Extensible to Maitri-II & future Indian polar deployments